

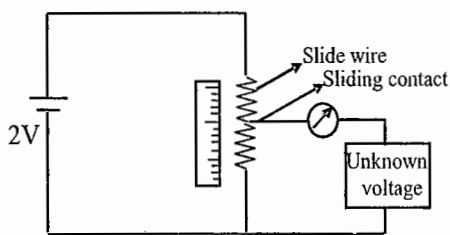
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Potentiometers & Instrument Transformers

Potentiometer or Null detector

It is an instrument designed to measure an unknown voltage by comparing it with a known voltage.

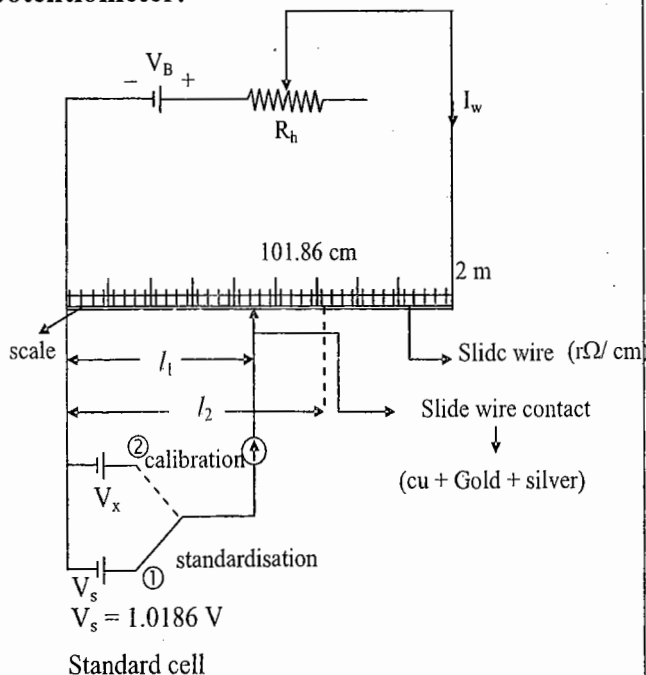
Basic potentiometer circuit:



It measures unknown voltage by comparing it with a known voltage or standard cell voltage.

Used to measure unknown voltages accurately

Circuit diagram of basic slide wire potentiometer:



$\ell \rightarrow$ length of slide wire of potentiometer.

$\ell_1 \rightarrow$ length of slide wire at which the galvanometer is balanced with standard cell voltage (V_s).

$\ell_2 \rightarrow$ length of slide wire at which the galvanometer is balanced with unknown EMF (V_x)

$I_w =$ working current

$$I_w = \frac{V_B}{(R_n + lr)}$$

Standardisation:

The process of adjusting working current (I_w), with the help of rheostat such that voltage drop across portion of slide wire is made equal to standard cell voltage.

Note: Standardisation is process of converting voltage in terms of centimeter scale

Standardisation calibration

$$V_s = I_w \cdot \ell_1 \cdot r$$

$$I_w = \frac{V_s}{\ell_1 \cdot r} \dots (1)$$

$$V_x = I_w \cdot \ell_2 \cdot r$$

$$I_w = \frac{V_x}{\ell_2 \cdot r} \dots (2)$$

By equation 1 and 2

$$V_x = V_s \frac{\ell_2}{\ell_1}$$

Note: Once standardization is done working current is kept constant.

Applications:

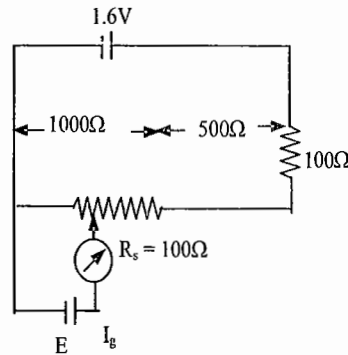
For more accuracy these are used for calibration of ammeter, voltmeter & watt meters.



Example 1:

An unknown voltage source E (with negligible internal resistance) is connected to a potentiometer circuit as shown in the figure below. If the galvanometer current $10\mu\text{A}$ with the direction as indicated, then value of E is

- (a) 0.9910V
- (b) 0.9991V
- (c) 1.0012V
- (d) 1.0048V



Sol: $I_g = 10\mu\text{A}$, $E = ?$

Write KVL for loop 1

$$-1.6 + (100 + 500)I_w + 1000(I_w + I_g) = 0$$

$$1600I_w + 1000I_g = 1.6$$

$$I_w = \frac{1.6 - 1 \times 10^{-3}}{1600} = 9.9375 \times 10^{-4} \text{ A}$$

Write KVL for loop 2

$$-E + 100 \times 10 \times 10^{-6} + 1000(10 \times 10^{-6} + 9.9375 \times 10^{-4}) = 0$$

$$E = 1.00475$$

- Practically in laboratories we are using Crompton's type of potentiometer.

Instrument Transformers

Transformers used in conjunction with measuring instruments for measurement purposes are called "Instrument transformers". The transformer used for measurement of current is called a "current transformer" or simply "CT". Transformers used for voltage measurements are called "voltage transformers" or "potential transformers" or simply "P. T" for short.

Ratio of Instrument Transformers:

1. Transformation Ratio:

$$R = \frac{\text{Primary wdg current}}{\text{secondary wdg current}} \quad \text{for a C. T.}$$

$$= \frac{\text{Primary wdg voltage}}{\text{secondary wdg voltage}} \quad \text{for a P. T.}$$

It is the ratio of the magnitude of the primary phasor to the secondary phasor.

2. Nominal Ratio:

It is the ratio of rated primary winding current (or voltage) to the rated secondary winding current (or voltage).

$$K_n = \frac{\text{rated primary winding current}}{\text{rated secondary winding current}} \quad \text{for a C. T.}$$

$$= \frac{\text{rated primary winding voltage}}{\text{rated secondary winding voltage}} \quad \text{for a P. T.}$$

3. Turns Ratio:

This is defined as

$$n = \frac{\text{number of turns of secondary winding}}{\text{number of turns of primary winding}} \quad \text{for a C. T.}$$

$$= \frac{\text{number of turns of primary winding}}{\text{number of turns of secondary winding}} \quad \text{for a P. T.}$$

The ratio errors are present in instrument transformers

$$(a) \% \text{ ratio error } (\% \sigma) = \frac{k_n - R}{R} \times 100$$

$$(b) \text{ Phase angle error } (\theta)$$

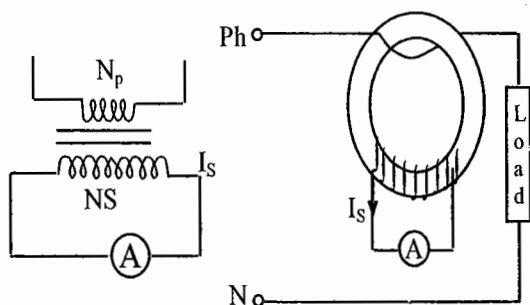
Ratio correction factor:

$$(R.c.f) = \frac{\text{Transformation ratio}(R)}{\text{No min al ratio}(k_n)}$$

The output of CT is called as burden.



Current Transformers:



C.T. primary is connected in series with the load current to be measured. Hence it is called a series T.I.F.

C.T secondary is short circuited by m.I Ammeter.

The ratio mentioned on C.T. name plate is a current ratio i.e. I_p/I_s .

Circular strip wound core is used for the C.T.

Note: C.T secondary should not be open circuited when it's primary is energized it's dangerous.

Reason: Secondary current is zero so secondary EMF is zero but to maintain same current in secondary terminals more magnetizing current is drawn because of this very high voltage is induced then insulation breakdown. It's dangerous to the person operating it.

Transformation ratio $(R) = \frac{I_p}{I_s}$

$$R = \frac{nI_s + I_0 \sin(\alpha + \delta)}{I_s} \quad (\because \delta = 0)$$

$$R = n + \frac{I_0 \sin \alpha}{I_s}$$

$$R = n + \frac{I_e}{I_s}$$

$$R = n \left[1 + \frac{I_e}{I_p} \right]$$

Phase angle error:

The primary and secondary phasor difference is to be 180° in case of ideal Transformer. But in practical this angle having a difference of θ is called phase angle error.

From the above phasor diagram.

$$\tan \theta = \frac{ab}{OC + cb} = \frac{I_0 \cos(\alpha + \delta)}{nI_s + I_0 \sin(\alpha + \delta)}$$

$$[\because I_0 \sin(\alpha + \delta) \ll nI_s]$$

$$\tan \theta \cong \frac{I_0 \cos \alpha}{nI_s} \quad (\because \delta \approx 0)$$

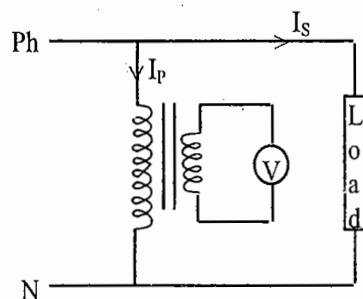
$$\therefore \theta = \frac{I_m}{nI_s} \text{ rad}$$

$$\theta = \frac{I_m}{nI_s} \times \frac{180}{\pi} \text{ degree}$$

Both ratio error and phase angle errors are due to the excitation current ' I_0 '.

Specifically the ratio error is due to the iron loss component ' I_e ' and phase angle error is due to the magnetizing component I_m .

Potential T.I.F:



P.T primary is connected in parallel with the load hence it is called as a shunt transformer.

The standard P.T. secondary voltage rating is 120V.



- P.T. secondary is open circuited by high resistance moving Iron voltmeter.
- It's basically a step down transformer.
- The effect of ratio error and phase error are less in P.T compared to C.T because it is involving with voltage measurement.
- For high voltage appreciation core type is used and for low type voltage, shell type core is used.
- Because cost point of view, potential transformers are not suitable for above 100kV voltage measurement. For this we are using (C.V.T) capacitive voltage transformers are used.

Actual ratio :

$$R = n + \frac{n_i (R_p \cos \Delta + X_p \sin \Delta) + I_e R_p + I_m X_p}{V_s}$$

$$\% \text{RatioError}(\sigma) = \frac{K_n - R}{R} \times 100$$

Phase angle error :

$$\theta = n + \frac{I_s (X_p \cos \Delta - R_p \sin \Delta) + I_e R_p \times I_m X_p}{n V_s} X_p$$

⇒ total resistance total reactance referred to primary side.

Difference between D.T and P.T.

D.T	P.T
1. in case of D.T (distribution transformer) is designed for changes in loads the voltage should be maintained constant i.e It depends voltage regulation.	1. Potential Transformer takes care of voltage ratio Between the primary and secondary.

2. we require η is main criteria in case of distribution Transformer	2. Accuracy is main criteria in case of P.T
3. D.T uses rectangular core	3. T uses circular core
4. Overloading capacity of Distribution Transformer is 20 to 30 % of full load	4. overloading capacity of potential is 2 to 3 % of full load

Example 2:

The constant for 3phase 3element integrating wattmeter is 0.15 revolution of disc per kwh, if the meter is normally used with a potential transformer of ratio 33000/110V and the current transformer of ratio 1000/10A, find the error expressed as a percentage of the correct reading from the following test figures for the instrument only:

Line voltage = 100V; Current 6.25A ; Power factor = 0.9 Time to complete 40 revolutions = 66s

- (a) -59.27% low (b) -29.28% low
(c) -63.32% low (d) -73.69% low

Ans: (a)

Sol: Actual energy consumed during the test period

$$= \sqrt{3} \times \text{ratio of P.T} \times \text{ratio of C.T} \times V_s I_s \cos \phi \times t$$

$$= \sqrt{3} \times \frac{33000}{110} \times \frac{1000}{10} \times 110 \times 6.25 \times \cos 0.9 \times \frac{66}{3600} \times 10^{-3}$$

$$= 654.85 \text{ kWh}$$

Energy recorded by meter during the test period

$$= \frac{40}{0.15} = 266.66 \text{ KWh}$$

$$\therefore \text{Percentage error} = \frac{266.66 - 654.85}{654.85} \times 100$$

$$= -59.27\% \text{ low}$$